

Secure Access

AI-Powered Smart Security & Access Monitoring System

Real-Time Face Recognition with Intelligent Alerting & AI Scene Analysis

■ TEAM CODERS

■ ASCALTHON 2026 – 24-HOUR ONLINE HACKATHON

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The Security Challenge



Unauthorized access

Restricted areas breached by unauthorised individuals



Manual attendance

Time-consuming, error-prone verification processes



Delayed response

Incidents detected too late for timely action



Human-dependent monitoring

Fatigue and oversight reduce reliability



Lack of automation

No intelligent automated threat triage or alerts



There is a need for an intelligent AI-powered real-time security monitoring platform.

Our Solution

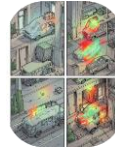
Secure Access — one intelligent platform that combines real-time CV, AI scene understanding, and instant notifications.



**Live Face
Recognition**



**Smart Access
Monitoring**



**AI Scene
Analysis**



**Telegram
Alerts**



Email Alerts



**Live
Dashboard &
Analytics**



**Student
Registration**



**Camera
Enrollment**



**Unauthorize
d Gallery**

System Architecture



End-to-end pipeline: real-time video → computer vision → decisioning → instant notifications → forensic storage and analytics.

Technology Stack



Frontend

React.js · HTML5 · CSS3 · JavaScript



Backend

Flask · Python



Artificial Intelligence

OpenCV · Face Recognition · AI Scene Analysis



Database

MongoDB · SQLite



Notifications

Telegram Bot API · SMTP Email



Developer Tools

GitHub · VS Code

Key Features

**Live Face
Recognition**

**Bounding Box
Detection**

**Authorized
Detection**

**Unauthorized
Detection**

**Student
Registration**

**Camera
Enrollment**

Live Camera Feed

AI Observation

**Dashboard
Analytics**

**Unauthorized
Gallery**

Telegram Alerts

Email Alerts

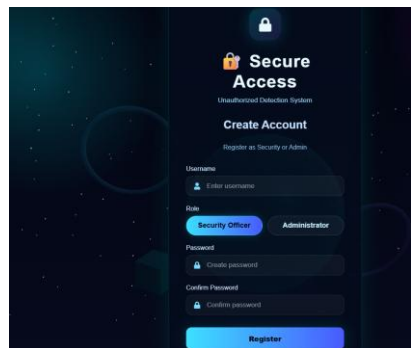
Role-Based Login

Complete Workflow

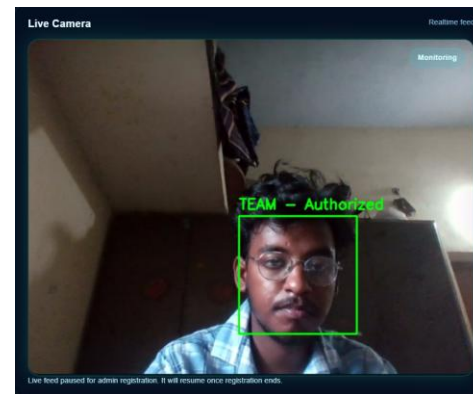
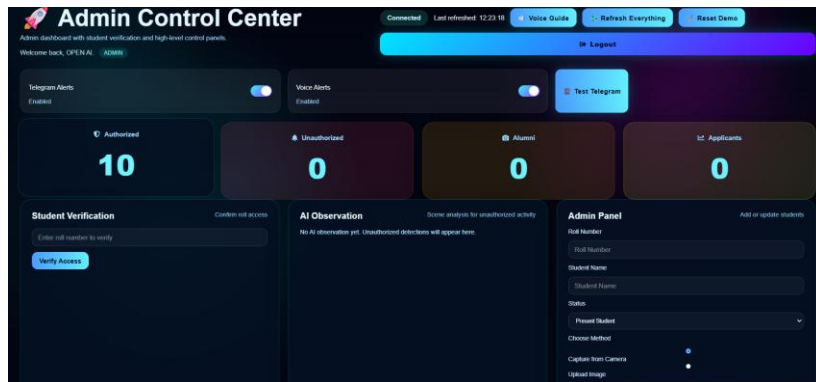


Elegant, gatekeeper logic with immediate alerting and forensic capture for every unknown event.

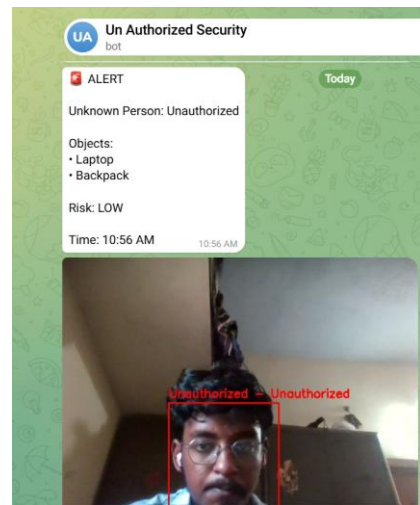
Live Demonstration – Placeholders



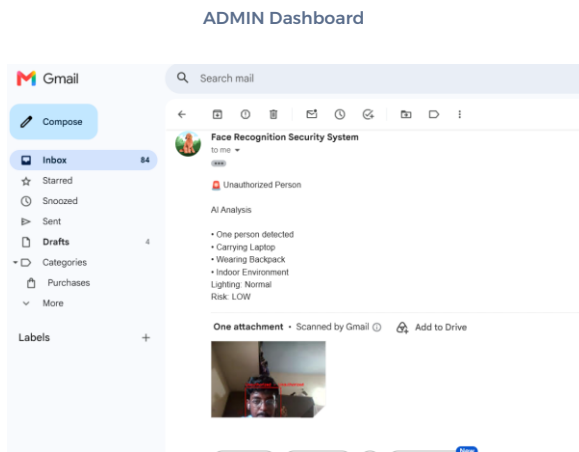
Login Page



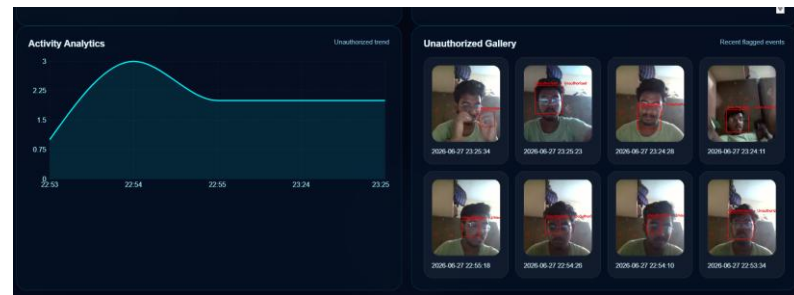
Live Camera Feed



Telegram Alert



Email Alert



Unauthorized Gallery

Project Achievements



Live Face Recognition

Real-time identity matching using OpenCV and face embeddings



Real-Time Camera Monitoring

Continuous stream ingestion and processing pipeline



AI Scene Analysis

Contextual object and activity detection for richer alerts



Dashboard Analytics

Forensic timeline, access logs, and visual insights



Student Registration

Secure enrollment flow with profile capture



Admin Camera Enrollment

Device onboarding and status monitoring



Telegram & Email Notifications

Immediate incident delivery to admins



Unauthorized Gallery

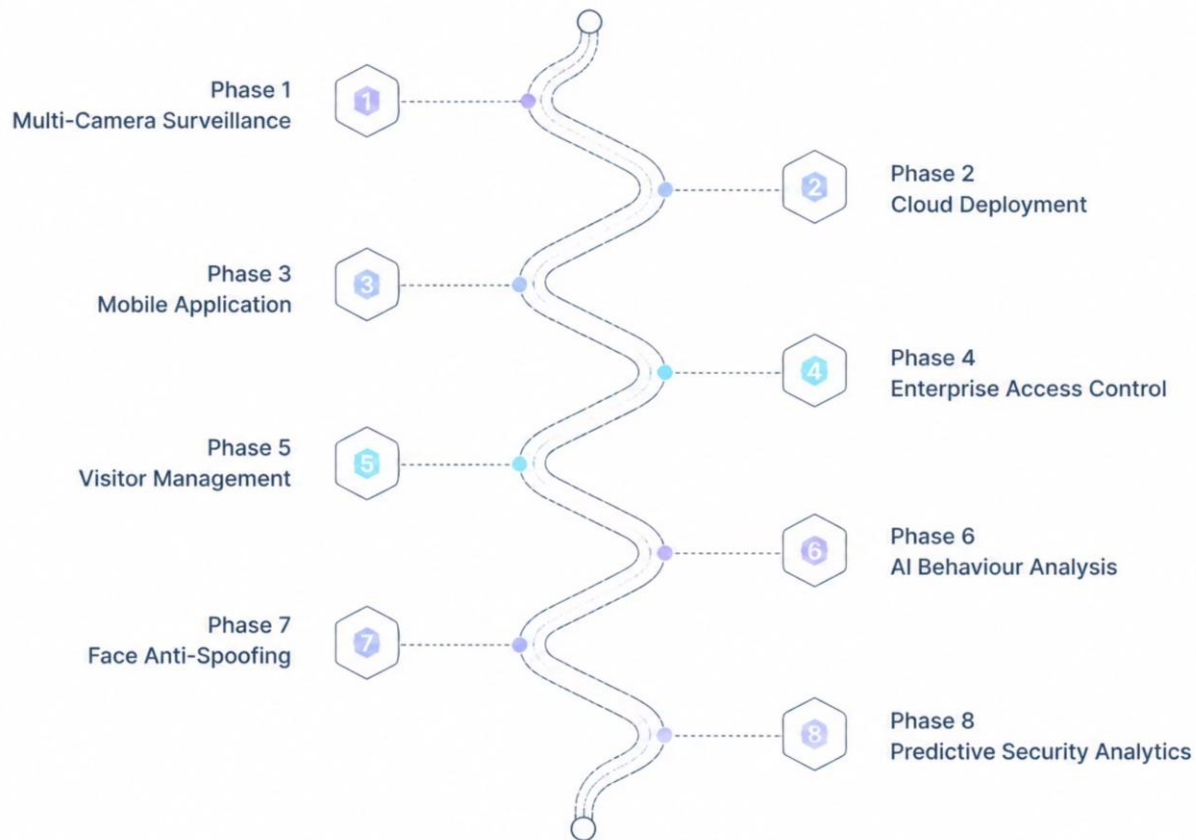
Sorted evidence for review and escalation



Role-Based Authentication

Tiered access for admins, operators and auditors

Future Roadmap



Phased rollout focusing on scale, robustness, mobile reach and advanced AI capabilities.

THANK YOU ❤️

THANK YOU FOR UR VALUABLE TIME AND ATTENTION

QUESTIONS & DISCUSSIONS

